TASK 12 – Re-evaluate and Adapt the Design

Decision: MODIFY the original design

A. Impact Analysis – 2 Marks

The metro introduces three changes:

1. A fourth zone, Airport, with a fare of Rs 60.
2. A 10% weekend discount that does not combine with the peak surcharge.
3. A monthly pass allowing unlimited journeys in two selected zones.

The following original components are affected:

Original component	**Required modification**
Fare constants	Add Airport fare = Rs 60
Zone enum	Add Airport
Fare switch statement	Add Airport case or replace with a fare table
Peak surcharge expression	Retain 20% surcharge
Fare calculation function	Add weekend discount priority

SmartCard class	Support journeys covered by a monthly pass
New MonthlyPass class	Store two selected zones and pass validity

Discount rule: The weekend discount replaces the peak surcharge. Therefore, a weekend journey during peak hours receives only the 10% weekend discount.

B. Evaluation of the Original Design – 2 Marks

1. Switch-based fare logic

The original switch-based fare calculation can support Airport by adding another case. However, adding more zones and fare rules makes the switch statement longer and harder to maintain.

A fare table is more flexible because new zones can be added by extending the table.

2. SmartCard class

The original SmartCard class can manage balance, recharge and fare deduction. However, it does not independently represent a monthly pass.

A separate MonthlyPass class is more suitable for storing two selected zones and checking whether a journey is covered.

3. Worked examples

Journey	**Calculation**	**Final fare**
Inner, weekday off-peak	Rs 10	Rs 10.00

Middle, weekday peak	20 + 20%	Rs 24.00
Airport, weekday peak	60 + 20%	Rs 72.00
Airport, weekend off-peak	60 − 10%	Rs 54.00
Airport, weekend peak	60 − 10% only	Rs 54.00
Monthly pass, selected zone	Unlimited covered journey	Rs 0.00

These examples demonstrate why weekend discount must take priority over peak surcharge.

C. Adaptation and Revised Design – 2 Marks

Revised design

I recommend modifying the original design using:

- A fare table containing four zone fares.
- Integer paise (long long) for accurate monetary calculations.
- A separate calculateFare() function.
- A MonthlyPass class for two selected zones.
- The existing SmartCard class for balance management.

Expected output

```
====================================
  REVISED METRO SMART CARD SYSTEM
====================================
```

FARE TABLE

Inner Rs. 10.00

Middle Rs. 20.00

Outer Rs. 30.00

Airport Rs. 60.00

FARE CALCULATION TESTS

Inner weekday off-peak: Rs. 10

Middle weekday peak: Rs. 24

Airport weekday peak: Rs. 72

Airport weekend off-peak: Rs. 54

Airport weekend peak: Rs. 54

MONTHLY PASS TESTS

Inner journey: Covered by monthly pass.

Airport journey: Covered by monthly pass.

Middle journey: Fare Rs. 24.00 deducted.

Remaining card balance: Rs. 76.00

PASS EXPIRY TEST

Inner journey: Fare Rs. 10.00 deducted.

Final card balance: Rs. 66.00

Note: The implementation demonstrates pass activation and deactivation. A production system must also check the actual monthly start and expiry dates.

D. Evidence-Based Defence – 2 Marks

1. Test results

Test case	Expected result	Actual result
Inner weekday off-peak	Rs 10	Rs 10
Middle weekday peak	Rs 24	Rs 24
Airport weekday peak	Rs 72	Rs 72
Airport weekend off-peak	Rs 54	Rs 54
Airport weekend peak	Rs 54	Rs 54
Journey in first pass zone	Free	Free
Journey in second pass zone	Free	Free
Journey outside pass zones	Normal fare	Rs 24
Journey after pass deactivation	Normal fare	Rs 10

2. Complexity analysis

Let z represent the number of zones.

Operation	Time complexity	Space complexity
Fare lookup	O(1)	O(z) for fare table
Fare calculation	O(1)	O(1)
Monthly pass zone checking	O(1)	O(1)
Balance deduction	O(1)	O(1)

3. Final design defence

The original design was suitable for three fixed zones, but the new requirements make a simple switch-based approach less flexible. Therefore, I recommend modifying the original design instead of completely replacing it.

The revised design provides:

- A flexible fare table for adding or changing zones.
- Accurate monetary calculations using integer paise.
- Separate weekend and peak rules to prevent combining discounts.
- A MonthlyPass class for unlimited journeys in two selected zones.
- A SmartCard class that continues to protect the balance and enforce the Rs 50 minimum balance rule for paid journeys.

The main trade-off is the additional complexity of maintaining multiple classes and fare rules. A real metro system must also validate pass expiry dates, handle simultaneous transactions and synchronize fare updates across all gates.

Final Recommendation: Modify and extend the original object-oriented design. It supports the fourth zone, weekend discount and monthly pass while maintaining the safety and accuracy of the original SmartCard system.